

Ticket #004: Disk Space Exhaustion - Incident Response

Symptoms: System monitoring alerts indicate critically low available disk space on the server. Some services are experiencing slow performance or intermittent failures when attempting to write data. Users report certain applications failing to save files or generate logs. Automated backups and scheduled tasks may be failing due to insufficient storage. System alerts indicate disk utilization approaching or exceeding capacity.

Resolution:

1. First, start with inspecting the content stored on the server's disk.

```
jay@mainserver:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	371M	1.3M	370M	1%	/run
efivarfs	256K	19K	238K	8%	/sys/firmware/efi/efivars
/dev/mapper/ubuntu--vg-ubuntu--lv	9.8G	9.8G	0	100%	/
tmpfs	1.9G	0	1.9G	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/vda2	1.9G	197M	1.6G	11%	/boot
/dev/vda1	1.1G	6.4M	1.1G	1%	/boot/efi
tmpfs	371M	12K	371M	1%	/run/user/1000

Analysis: The filesystem ubuntu--vg-ubuntu--lv has its mount point at the root directory and is at full capacity. Ubuntu--vg-ubuntu--lv is managed by the Logical Volume Manager so expanding the volume is possible if needed.

2. Narrowing down to which directory in ubuntu--vg-ubuntu--lv is consuming the most space.

```
jay@mainserver:~$ sudo du -sh /* 2>/dev/null
```

0	/bin
4.0K	/bin.usr-is-merged
203M	/boot
4.0K	/cdrom
0	/dev
6.2M	/etc
168K	/home
0	/lib
4.0K	/lib.usr-is-merged
16K	/lost+found
4.0K	/media
4.0K	/mnt
4.0K	/opt
0	/proc
40K	/root
34M	/run
0	/sbin
4.0K	/sbin.usr-is-merged
8.0K	/snap
4.0K	/srv
1.9G	/swap.img
0	/sys
1.8G	/tmp
2.6G	/usr
3.6G	/var

Analysis: The most suspicious directory is /var, using 3.6 GB. Since it stores dynamic data, it is a common source of disk growth and is likely contributing to the root filesystem reaching capacity.

3. Moving from directory-level to file-level investigation in the /var directory. Breaking /var down into its major components.

```
[jay@mainserver:~$ sudo du -sh /var/*
[[sudo] password for jay:
1.7M    /var/backups
109M    /var/cache
4.0K    /var/crash
291M    /var/lib
4.0K    /var/local
0       /var/lock
3.3G    /var/log
4.0K    /var/mail
4.0K    /var/opt
0       /var/run
4.0K    /var/snap
16K     /var/spool
48K     /var/tmp
12K     /var/www
```

Analysis: /var/log is the directory containing the most data.

4. Inspecting disk usage on /var/log.

```
[jay@mainserver:~$ sudo du -sh /var/log/*
28K     /var/log/alternatives.log
1.5G    /var/log/app
0       /var/log/apport.log
4.0K    /var/log/apport.log.1
4.0K    /var/log/apport.log.2.gz
132K    /var/log/apt
24K     /var/log/auth.log
136K    /var/log/auth.log.1
24K     /var/log/auth.log.2.gz
60K     /var/log/bootstrap.log
4.0K    /var/log/btmp
72K     /var/log/cloud-init.log
8.0K    /var/log/cloud-init-output.log
4.0K    /var/log/dist-upgrade
48K     /var/log/dmesg
48K     /var/log/dmesg.0
```

Analysis: The largest entry is /var/log/app at 1.5 GB.

5. Listing the log files in the /var/log/app.

```
[jay@mainserver:~$ sudo ls -lh /var/log/app
total 1.5G
-rw-r--r-- 1 root root 1.5G Mar 16 15:12 payment-service.log
```

Analysis: The root cause is the payment-service.log file.

6. Inspecting the payment-service.log file and looking for repeated errors, spammy logs, debug output, and looping messages.

```
jay@mainserver:~$ sudo tail -n 50 /var/log/app/payment-service.log
jay@mainserver:~$ sudo head /var/log/app/payment-service.log
jay@mainserver:~$ sudo file /var/log/app/payment-service.log
/var/log/app/payment-service.log: data
jay@mainserver:~$
```

Analysis: The file payment-service.log is in a raw binary format, which explains why no readable output appears in either the tail or head commands.

7. Verifying if the file is safe to clear by checking whether it is actively used by a service.

```
jay@mainserver:~$ sudo lsof | grep payment-service
jay@mainserver:~$
```

Analysis: No active processes are found holding an open file handle to the log. The file likely grew due to prior logging activity and it is safe to clear.

8. Clearing payment-service.log file to 0 bytes with truncate and verifying with df -h.

```
jay@mainserver:~$ sudo truncate -s 0 /var/log/app/payment-service.log
[sudo] password for jay:
jay@mainserver:~$ df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	371M	1.3M	370M	1%	/run
efivarfs	256K	19K	238K	8%	/sys/firmware/efi/efivars
/dev/mapper/ubuntu--vg-ubuntu--lv	9.8G	8.4G	948M	90%	/
tmpfs	1.9G	0	1.9G	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/vda2	1.9G	197M	1.6G	11%	/boot
/dev/vda1	1.1G	6.4M	1.1G	1%	/boot/efi
tmpfs	371M	12K	371M	1%	/run/user/1000

```
jay@mainserver:~$
```

Analysis: Truncating the file reduced disk usage on /dev/mapper/ubuntu--vg-ubuntu--lv to 90%. There is still another file consuming a large amount of disk space.

9. Investigating what other file in /dev/mapper/ubuntu--vg-ubuntu--lv is taking up excessive disk space.

```
400M    /var/log/syslog.2
188K    /var/log/syslog.2.gz
400M    /var/log/syslog.3
400M    /var/log/syslog.4
400M    /var/log/syslog.5
```

Analysis: These are the largest log files in /var/log and are old rotated logs, so deleting them is safe.

10. Deleting old rotated log files.

```
jay@mainserver:~$ sudo rm /var/log/syslog.2 /var/log/syslog.2.gz
jay@mainserver:~$ sudo rm /var/log/syslog.3 /var/log/syslog.4
jay@mainserver:~$ sudo rm /var/log/syslog.5
```

11. Checking disk usage of ubuntu--vg-ubuntu--lv

Filesystem	Size	Used	Avail	Use%	Mounted on
tmpfs	371M	1.3M	370M	1%	/run
efivarfs	256K	19K	238K	8%	/sys/firmware/efi/efivars
/dev/mapper/ubuntu--vg-ubuntu--lv	9.8G	7.0G	2.3G	76%	/
tmpfs	1.9G	0	1.9G	0%	/dev/shm
tmpfs	5.0M	0	5.0M	0%	/run/lock
/dev/vda2	1.9G	197M	1.6G	11%	/boot
/dev/vda1	1.1G	6.4M	1.1G	1%	/boot/efi
tmpfs	371M	12K	371M	1%	/run/user/1000

Analysis: ubuntu--vg-ubuntu--lv remains high (76%) after log cleanup. /var/log has been reduced and is no longer a primary contributor.

12. The next step is to investigate the other directories.

1.9G	/swap.img
0	/sys
1.8G	/tmp
2.6G	/usr
860M	/var

Analysis: Largest directory is /usr but priority is given to /tmp (1.8 GB) since it consists of temporary, non-essential files and offers opportunity for immediate disk space recovery.

13. Moving a level deeper into the /tmp directory and breaking it down into its major components

```
[jay@mainserver:~$ sudo du -h /tmp
4.0K    /tmp/systemd-private-4b10d872bef84eaab9a4e7c19907f655-fwupd.service-2inkYe
4.0K    /tmp/systemd-private-4b10d872bef84eaab9a4e7c19907f655-systemd-logind.service-pRwpC3/tmp
8.0K    /tmp/systemd-private-4b10d872bef84eaab9a4e7c19907f655-systemd-logind.service-pRwpC3
1.8G    /tmp/batch-processing
4.0K    /tmp/.ICE-unix
4.0K    /tmp/snap-private-tmp
```

Analysis: /tmp/batch-processing is responsible for the majority of /tmp disk usage.

14. Using the command lsof to verify whether any processes have active file handles on batch-processing before performing cleanup.

```
[jay@mainserver:~$ lsof +D /tmp/batch-processing
lsof: WARNING: can't stat() tracefs file system /sys/kernel/debug/tracing
Output information may be incomplete.
```

Analysis: Output returned a warning related to the tracefs virtual filesystem, indicating it cannot fully inspect certain kernel tracing paths. This is expected behavior and unrelated to /tmp/batch-processing. No active file usage for /tmp/batch-processing was shown in the output. However, additional verification (e.g., fuser) is required before confirming it is safe to delete.

15. Using fuser for additional verification.

```
[jay@mainserver:~$ sudo fuser -v /tmp/batch-processing/
jay@mainserver:~$
```

Analysis: No active file usage for /tmp/batch-processing is shown in the output. It is safe to delete.

16. Deleting /tmp/batch-processing and checking disk percentage.

```
[jay@mainserver:~$ sudo rm -rf /tmp/batch-processing
[jay@mainserver:~$ sudo df -h
Filesystem                Size      Used Avail Use% Mounted on
tmpfs                     371M    1.3M   370M    1% /run
efivarfs                  256K     19K   238K    8% /sys/firmware/efi/efivars
/dev/mapper/ubuntu--vg-ubuntu--lv 9.8G    5.3G    4.0G   57% /
tmpfs                     1.9G         0    1.9G    0% /dev/shm
tmpfs                     5.0M         0    5.0M    0% /run/lock
/dev/vda2                 1.9G    197M    1.6G   11% /boot
/dev/vda1                 1.1G     6.4M    1.1G    1% /boot/efi
tmpfs                     371M     12K   371M    1% /run/user/1000
```

Analysis: The /dev/mapper/ubuntu--vg-ubuntu--lv filesystem has returned to a healthy level of disk usage. The disk pressure issue is successfully resolved.

Conclusion: The root filesystem reached critical disk usage, causing service degradation and write failures. Analysis identified excessive usage in /var/log, specifically a 1.5 GB payment-service.log file containing invalid binary data. After confirming no active file handles, the file was truncated. Additional space was recovered by removing old rotated logs. Further investigation revealed high usage in /tmp/batch-processing. After verifying no active processes were using the files, the directory was deleted. These actions restored disk usage to a healthy level and resolved the issue.